

Killian **QUETEL**

Mechanical & Surface Design Engineer · CATIA Expert

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PROFILE

CATIA Expert and mechanical engineer shaped by high-tech industries, from CAD software R&D to aerospace and space. I manage the portfolio of **CATIA core design roles** (Part, Surface and Sketcher), turning industrial needs into R&D roadmaps on themes such as **Body-in-White**, and I keep my hands on the product: high-quality surfacing, robust parametric and **modular design**, and simulation-driven engineering.

EXPERIENCE

CATIA R&D Core Engineering Roles Portfolio Manager — Apr 2024 – present

Dassault Systèmes · Vélizy-Villacoublay, France

- Own the portfolio and roadmap of the CATIA core design roles (**Part, Surface and Sketcher**) on 3DEXPERIENCE, from customer needs to prioritized R&D requirements.
- Define **LEO**, the **LLM-powered AI assistant** of 3DEXPERIENCE.
- Work with many customers across industries such as **automotive, aerospace, transportation & defense**, from their engineering processes to CATIA requirements.
- Build industrial demonstrators on real products: **F1 virtual twin** (front wings 2019 & 2025, brake air duct entry), **aircraft wing**, **car Body-in-White** (generative door and hood) and CubeSat.
- CATIA Talk speaker and demonstration videos for the official CATIA page.

Mechanical System Engineer (VIE via Alten) — Jul 2023 – Jan 2024

DCUBED · Munich, Germany

- Owned the deployment mechanisms of a multi flat-panel reflectarray antenna for small satellites (**ESA ARTES – MARAS project**).
- Covered every step: R&D, simulation & validation, manufacturing and test campaign; co-author of the related **IAC 2024** paper.

Mechanical Engineer & CAD Process Engineer (VIE via Alten) — 2022 – 2023

Airbus · Hamburg & Munich, Germany

- **A350 cabin:** modeling of galley sets in CATIA V5 (4 months).
- **MDO CAD process:** topology optimization support on 3DEXPERIENCE and export control compliance (6 months).

CATIA R&D Engineer (Apprenticeship) — Sep 2020 – Sep 2022

Dassault Systèmes · Vélizy-Villacoublay, France

- **Generative design:** demonstrators coupling structural and fluid topology optimization, subdivision modeling, city- and bridge-scale models by reverse engineering (Bir-Hakeim bridge: from generative design to steel structure).
- **Sustainable campus:** energy-harvesting mechanisms (speed bump, turnstile, piezo floor), simulated in SIMULIA and Dymola, with a 1:20 working 3D-printed model.

Production Quality Assurance (Apprenticeship) — 2019 – 2020

Safran Aircraft Engines · France

- Quality control on turbine-blade lines (M88 Rafale, LEAP, CFM56), reducing costs and lead times.

PERSONAL PROJECT



Formula 1 2026 · full car at scale 1:1

Designed from scratch in CATIA within the **2026 FIA technical regulations**: modular, GSD+ surfaces only, aero validated with SIMULIA CFD.

300+ hours **60k+** features **0** B-Rep Oct 2025 – present

SKILLS

CAD & surfacing

CATIA 3DEXPERIENCE · Part Design · GSD+ · Sketcher · Assembly Design · Modular design · Imagine & Shape · Generative Design · CATIA V5 · SolidWorks · Fusion 360

Simulation

SIMULIA (CFD, structural) · Abaqus · Ansys · HyperMesh · OptiStruct · Dymola

Product & management

Portfolio & roadmap · Requirements · Agile (LeSS) · Customer demos

AI

LLM assistants · AI agents & prompting · AI-assisted design · AI-assisted coding

Code & tools

3D printing · Python · C# · MATLAB · Arduino · Adobe Suite

EDUCATION

MSc Engineering – ESILV
Computational mechanics & digital modeling · 2018 – 2022 · work-study

BSc Science – Paris-Saclay
1st year, materials & aerospace · 2017 – 2018

HIGHLIGHTS

IAC 2024, Milan – co-author
Design & deployment feasibility study of a 3-panel reflectarray antenna for a 12U CubeSat

CATIA Talk – speaker
Live LinkedIn presentation in English, 150+ viewers

3DEXPERIENCE certifications (2022)
Function Driven Generative Designer · Product Manager – Associate

LANGUAGES

French	●●●●●	Native
English	●●●●○	C1
Spanish	●●●○○	B1
German	●●○○○	A2

BEYOND WORK

- French Army reservist since 2019
- Running, marathons & trail · strength training · extreme sports · cycling
- Motorsport, photography, mountains & travel